

Portfolio Design System Guidelines

Color Palette

Primary Colors

CSS

```
/* Main Brand Gradients */
bg-gradient-to-br from-indigo-600 via-purple-600 to-blue-700
bg-gradient-to-r from-blue-600 to-purple-600
bg-gradient-to-br from-slate-900 via-blue-900 to-indigo-900
bg-gradient-to-br from-slate-50 via-blue-50 to-indigo-50

/* Accent Colors */
text-yellow-400      /* Success/Highlight */
text-green-400       /* Available/Positive */
text-pink-400        /* Creative/Emphasis */
text-blue-400        /* Links/Interactive */
text-purple-400      /* Secondary accent */
text-cyan-400        /* Tech/Modern */
text-orange-400      /* Warning/Energy */
```

Neutral Colors

CSS

```
/* Light Theme Backgrounds */
bg-slate-50      /* Light sections */
bg-slate-100     /* Subtle backgrounds */
bg-white         /* Cards/containers */
bg-gray-50       /* Alternative light */

/* Dark Theme Backgrounds */
bg-slate-900     /* Dark sections */
bg-slate-800     /* Dark containers */
bg-gray-900      /* Alternative dark */

/* Text Colors */
text-slate-800   /* Primary dark text */
text-slate-600   /* Secondary text */
text-slate-400   /* Muted text */
text-white       /* Light text on dark */
text-blue-100    /* Light text on gradients */
text-slate-300   /* Medium light text */
```

Typography Scale

Headings

CSS

```
/* Hero/Display Titles */
text-4xl md:text-6xl lg:text-7xl font-bold

/* Section Headings */
text-3xl md:text-5xl font-bold

/* Subsection Headings */
text-2xl md:text-3xl font-bold

/* Card/Component Titles */
text-xl md:text-2xl font-bold

/* Small Headings */
text-lg font-bold
```

Body Text

CSS

```
/* Large Body Text */
text-lg md:text-xl leading-relaxed

/* Regular Body Text */
text-base leading-relaxed

/* Small Text */
text-sm

/* Extra Small Text */
text-xs
```

Font Weights

CSS

font-bold	/* Headings, CTAs */
font-semibold	/* Subheadings, emphasis */
font-medium	/* Navigation, labels */
font-normal	/* Body text */

Spacing & Layout

Section Spacing

CSS

```
/* Large sections */
```

```
py-24 px-8
```

```
/* Medium sections */
```

```
py-16 px-8
```

```
/* Small sections */
```

```
py-12 px-8
```

```
/* Container max-width */
```

```
max-w-6xl mx-auto /* Standard container */
```

```
max-w-7xl mx-auto /* Wide container */
```

```
max-w-4xl mx-auto /* Narrow container */
```

Component Spacing

CSS

```
/* Between elements */
```

```
space-y-8 /* Large gaps */
```

```
space-y-6 /* Medium gaps */
```

```
space-y-4 /* Small gaps */
```

```
space-y-2 /* Tiny gaps */
```

```
/* Internal padding */
```

```
p-8 /* Large cards */
```

```
p-6 /* Medium cards */
```

```
p-4 /* Small cards */
```

```
p-3 /* Buttons */
```

Grid Systems

CSS

```
/* Responsive grids */
grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-8
grid grid-cols-1 lg:grid-cols-2 gap-12
grid md:grid-cols-2 lg:grid-cols-4 gap-8

/* Flex layouts */
flex flex-col sm:flex-row gap-4
flex items-center justify-between
flex flex-wrap gap-4
```

Component Patterns

Cards

CSS

```
/* Standard Card */
bg-white/80 backdrop-blur-sm p-8 rounded-2xl shadow-xl border border-white/50

/* Dark Card */
bg-white/10 backdrop-blur-md p-8 rounded-2xl shadow-2xl border border-white/20

/* Hover Effects */
hover:shadow-2xl hover:scale-105 transition-all duration-300
```

Buttons

Primary Button

CSS

```
bg-gradient-to-r from-blue-600 to-purple-600 text-white px-8 py-4 rounded-full font-bold
```

Secondary Button

CSS

```
bg-transparent border-2 border-white/50 text-white px-8 py-4 rounded-full font-bold text-white
```

Ghost Button

CSS

```
bg-white/20 backdrop-blur-sm px-6 py-3 rounded-full text-white font-medium hover:bg-white
```

Badges/Tags

```
CSS

/* Status Badge */
bg-green-500/20 border border-green-400/30 px-6 py-3 rounded-full backdrop-blur-sm

/* Tech Stack Tag */
bg-white/20 text-white px-3 py-1 rounded-full text-xs font-medium

/* Service Tag */
bg-indigo-100 text-indigo-800 px-4 py-2 rounded-full text-sm font-medium
```

Animation Patterns

Framer Motion Variants

```
javascript

// Fade in from bottom
const fadeInUp = {
  initial: { opacity: 0, y: 60 },
  animate: { opacity: 1, y: 0 },
  transition: { duration: 0.8, ease: "easeOut" }
};

// Slide in from left
const slideInLeft = {
  initial: { opacity: 0, x: -50 },
  animate: { opacity: 1, x: 0 },
  transition: { duration: 0.6 }
};

// Stagger children
const staggerContainer = {
  animate: {
    transition: {
      staggerChildren: 0.1
    }
  }
};

// Scale on hover
whileHover={{ scale: 1.05, y: -2 }}
whileTap={{ scale: 0.95 }}
```

CSS Animations

CSS

```
/* Pulse effect */
```

```
animate-pulse
```

```
/* Bounce effect */
```

```
animate-bounce
```

```
/* Custom transitions */
```

```
transition-all duration-300
```

```
transition-colors duration-200
```

```
transition-transform duration-300
```

Background Patterns

Gradient Backgrounds

CSS

```
/* Hero sections */
```

```
bg-gradient-to-br from-indigo-600 via-purple-600 to-blue-700
```

```
/* Light sections */
```

```
bg-gradient-to-br from-slate-50 via-blue-50 to-indigo-50
```

```
/* Dark sections */
```

```
bg-gradient-to-br from-slate-900 via-blue-900 to-indigo-900
```

Glassmorphism Effects

CSS

```
/* Light glass */
```

```
bg-white/80 backdrop-blur-sm border border-white/50
```

```
/* Dark glass */
```

```
bg-white/10 backdrop-blur-md border border-white/20
```

```
/* Strong glass */
```

```
bg-white/20 backdrop-blur-lg border border-white/30
```

Floating Elements

CSS

```
/* Decorative blurs */
absolute top-1/4 left-1/4 w-96 h-96 bg-blue-500/10 rounded-full blur-3xl

/* Animated floating */
animate-pulse delay-1000
```

Responsive Design

Breakpoint Strategy

CSS

```
/* Mobile First Approach */
/* Base: 0-639px */
sm: /* 640px+ */
md: /* 768px+ */
lg: /* 1024px+ */
xl: /* 1280px+ */
2xl: /* 1536px+ */
```

Common Responsive Patterns

CSS

```
/* Text scaling */
text-4xl md:text-6xl lg:text-7xl

/* Grid responsive */
grid-cols-1 md:grid-cols-2 lg:grid-cols-3

/* Padding responsive */
px-4 md:px-8 lg:px-12

/* Flex responsive */
flex-col sm:flex-row

/* Hidden/shown responsive */
hidden lg:block
block lg:hidden
```

Interactive States

Hover Effects

CSS

```
/* Scale up */  
hover:scale-105  
  
/* Shadow enhancement */  
hover:shadow-2xl  
  
/* Color changes */  
hover:text-blue-400  
hover:bg-white/30  
  
/* Transform */  
hover:translate-y-2
```

Focus States

CSS

```
focus:outline-none focus:ring-2 focus:ring-blue-500 focus:border-transparent
```

Active States

CSS

```
active:scale-95  
active:bg-blue-700
```

Icon Guidelines

Icon Sizes

CSS

```
w-4 h-4    /* Small icons */  
w-5 h-5    /* Medium icons */  
w-6 h-6    /* Large icons */  
w-8 h-8    /* Extra large icons */
```

Icon Colors

CSS

```
text-slate-400    /* Muted icons */
text-white         /* Light icons */
text-blue-400     /* Accent icons */
currentColor      /* Inherit parent color */
```

Content Guidelines

Section Structure

1. **Header with badge** (optional)
2. **Main heading** (large, bold)
3. **Subtitle/description** (medium, muted)
4. **Content/grid**
5. **Call-to-action** (if applicable)

Card Structure

1. **Icon/visual element**
2. **Title** (bold)
3. **Description** (muted)
4. **Tags/badges** (if applicable)
5. **Action button/link** (if applicable)

Animation Timing

- **Entrance animations:** 0.6-0.8s duration
 - **Hover effects:** 0.2-0.3s duration
 - **Stagger delay:** 0.1-0.2s between elements
 - **Page transitions:** 0.5s duration
-

Accessibility Guidelines

Color Contrast

- Ensure minimum 4.5:1 contrast ratio for text
- Use multiple visual cues (not just color)
- Test with color blindness simulators

Interactive Elements

- Minimum 44px touch target size
- Clear focus indicators
- Semantic HTML elements
- ARIA labels where needed

Typography

- Readable font sizes (minimum 16px)
 - Sufficient line height (1.5-1.6)
 - Reasonable line length (45-75 characters)
-

Performance Guidelines

Animation Performance

- Use `transform` and `opacity` for animations
- Avoid animating `width`, `height`, `top`, `left`
- Use `will-change` sparingly
- Prefer CSS transforms over JavaScript

Image Optimization

- Use WebP format when possible
 - Implement lazy loading
 - Provide appropriate alt text
 - Use responsive images
-

Component Checklist

When creating new components, ensure:

- ☐ **Responsive design** across all breakpoints
- ☐ **Consistent spacing** using design system values
- ☐ **Proper color usage** from defined palette
- ☐ **Smooth animations** with appropriate timing
- ☐ **Accessibility** compliance
- ☐ **Hover/focus states** for interactive elements
- ☐ **Loading states** if applicable
- ☐ **Error states** if applicable
- ☐ **Clean, semantic HTML**
- ☐ **Performance optimization**

Example Component Template

```
jsx

import React from 'react';
import { motion } from 'framer-motion';

const ComponentName = () => {
  const fadeInUp = {
    initial: { opacity: 0, y: 60 },
    animate: { opacity: 1, y: 0 },
    transition: { duration: 0.8, ease: "easeOut" }
  };

  return (
    <section className="py-24 px-8 bg-gradient-to-br from-slate-50 via-blue-50 to-indigo-50">
      <div className="max-w-6xl mx-auto">
        {/* Header */}
        <motion.div
          variants={fadeInUp}
          initial="initial"
          whileInView="animate"
          viewport={{ once: true }}
          className="text-center mb-16">
          >
          <h2 className="text-3xl md:text-5xl font-bold mb-6 text-slate-800">
            Section Title
          </h2>
          <p className="text-xl text-slate-600 max-w-3xl mx-auto leading-relaxed">
            Section description
          </p>
        </motion.div>

        {/* Content */}
        <div className="grid md:grid-cols-2 lg:grid-cols-3 gap-8">
          {/* Component content */}
        </div>
      </div>
    </section>
  );
};

export default ComponentName;
```

This design system ensures consistency, accessibility, and professional quality across all components in your portfolio.